

TB XYZ Panner

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Overview

Places a sound left or right, forward or back, high or low, and near or far.

What this plug-in solves

Conventional pan controls only left/right gain. TB XYZ Panner keeps that reliable foundation and adds procedural spectral, level, and reflection cues for height, distance, and front/back perspective. Headphones mode adds KEMAR-based binaural rendering, while surround layouts use speaker azimuth remapping.

What you can expect

- Reliable stereo left/right placement
- Procedural near/far and up/down cues
- Front/back impression without rewriting pan
- Headphone binaural and multichannel positioning options

How it works

TB XYZ Panner combines level, stereo balance, filtering, and room cues to create the impression that a sound occupies a position around the listener. The axes are perceptual controls: left/right establishes lateral position, front/back and distance change closeness, and height adds vertical contrast where the listening setup can support it.

Place the sound with the main axes first, then add room and distance cues to connect it to the scene. Render Mode adapts the result for the intended speaker or headphone presentation. Always check the position on more than one playback system because extreme width or height cues can translate differently.

Quick start

- 1 Choose Stereo for speakers or Headphones for binaural monitoring.
- 2 Set Pan in Top View or with the Pan control.
- 3 Set Front/Back vertically in Top View.
- 4 Set Height and Distance in Side View.

- 5 Add Room only as needed.
- 6 Trim Output and verify on the actual delivery format.

Interface and key sections



This is a direct capture of the current plug-in interface. Use the visible labels to locate the areas and controls explained below.

Top View

Horizontal movement controls Pan; vertical movement controls Front/Back. Top View does not change Distance.

Side View

Horizontal movement controls Distance and vertical movement controls Height.

PanCore

Pan ranges from hard left through center to hard right. It remains the stable localization foundation and is not directly rewritten by front/back cues.

Height

Uses tonal and ceiling/floor reflection balance to suggest elevation. It is a psychoacoustic cue rather than a physical vertical speaker position in stereo.

Distance and Room

Distance combines attenuation, low-pass damping, and early reflections. Room controls the reflection contribution. The direct path is not simply moved by a modulated delay.

Render Mode

Stereo is intended for speaker-compatible panning. Headphones engages a compact KEMAR HRIR-based HRTF renderer; individual listeners may perceive front/back differently.

Surround and Output

On supported 5.1/7.1 outputs, non-LFE channels are remapped by speaker azimuth. Output Trim preserves headroom after spatial processing.

Controllable properties

The guide uses the names shown on the plug-in panel. Each explanation describes the audible result, a useful way to approach the control, and an important interaction to listen for.

Control	What it does and when to use it
Bypass	Turns the complete effect off or on for a direct before-and-after comparison. Compare with bypass at a similar loudness. If the effect creates a tail, let that tail settle before judging the difference.
Distance	Moves the sound from close to far away. Check the result on speakers, headphones, and in mono when possible. Extreme spatial settings can feel exciting but may not translate everywhere.
Front Back	Moves the sound in front of or behind the listener. Check the result on speakers, headphones, and in mono when possible. Extreme spatial settings can feel exciting but may not translate everywhere.
Height	Moves the sound below or above the listener. Check the result on speakers, headphones, and in mono when possible. Extreme spatial settings can feel exciting but may not translate everywhere.
Output Trim	Lowens the final level to leave headroom after spatial processing. Match the final loudness before deciding whether the processing sounds better. Extra level can make almost any setting seem more impressive.
Pan	Moves the sound left or right. Check the result on speakers, headphones, and in mono when possible. Extreme spatial settings can feel exciting but may not translate everywhere.
Program	Loads a factory starting point. Adjust the controls afterward to fit the current sound. Use a preset as a direction, not a finished answer. Change one section at a time so it is clear which adjustment improves the source.
Render Mode	Chooses regular stereo speakers or headphone-focused spatial rendering. Check the result on speakers, headphones, and in mono when possible. Extreme spatial settings can feel exciting but may not translate everywhere.
Room	Adds surrounding room cues so the position feels less dry and close. Check the result on speakers, headphones, and in mono when possible. Extreme spatial settings can feel exciting but may not translate everywhere.

Practical uses

Near overhead object

- Use a centered or slightly offset Pan.
- Raise Height in Side View.
- Keep Distance low and use modest Room.

Expected result: The source stays direct while gaining an upward cue.

Far rear ambience

- Move Front/Back toward rear in Top View.
- Increase Distance in Side View.
- Raise Room and reduce Output if the field builds up.

Expected result: The source feels farther and less direct with a rearward impression.

Common pitfalls

Exact rear and height localization on two speakers is physically limited. Reduce the main amount, feedback, drive, or input first, then rebuild the sound while watching the output meter and listening after the source stops.

The HRTF is not personalized, so front/back confusion is possible. Return the strongest control toward neutral, compare with bypass at a similar loudness, and add the processing back one section at a time.

Always verify headphone work on speakers and vice versa when both are delivery targets. Reduce the main amount, feedback, drive, or input first, then rebuild the sound while watching the output meter and listening after the source stops.